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[News] MRDIMM/MCRDIMM to be the New Sought-Afterers in Memory Field

2024-07-29 | Semiconductors | editor | f in t



Please note that this article cites information from [Tom's Hardware](#), [Micron](#) and WeChat account [DRAMeXchange](#).

Amidst the tide of artificial intelligence (AI), new types of DRAM represented by HBM are embracing a new round of development opportunities. Meanwhile, driven by server demand, MRDIMM/MCRDIMM have emerged as new sought-afterers in the memory industry, stepping onto the “historical stage.”

According to [a report from WeChat account DRAMeXchange](#), currently, the rapid development of AI and big data is boosting an increase in the number of CPU cores in servers. To meet the data throughput requirements of each core in multi-core CPUs, it is necessary to significantly increase the bandwidth of memory systems. In this context, HBM modules for servers, MRDIMM/MCRDIMM, have emerged.

- JEDEC Announces Details of the DDR5 MRDIMM Standard

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- **JEDEC Announces Details of the DDR5 MRDIMM Standard**

On July 22, [JEDEC announced that](#) it will soon release the DDR5 Multiplexed Rank Dual Inline Memory Modules (MRDIMM) and the next-generation LPDDR6 Compression-Attached Memory Module (CAMP) advanced memory module standards, and introduced key details of these two types of memory, aiming to support the development of next-generation HPC and AI. These two new technical specifications were developed by JEDEC's JC-45 DRAM Module Committee.

As a follow-up to JEDEC's JESD318 CAMP2 memory module standard, JC-45 is developing the next-generation CAMP module for LPDDR6, with a target maximum speed of over 14.4GT/s. In light of the plan, this module will also provide 24-bit wide subchannels, 48-bit wide channels, and support "connector array" to meet the needs of future HPC and mobile devices.

DDR5 MRDIMM supports multiplexed rank columns, which can combine and transmit multiple data signals on a single channel, effectively increasing bandwidth without additional physical connections. It is reported that JEDEC has planned multiple generations of DDR5 MRDIMM, with the ultimate goal of increasing its bandwidth to 12.8Gbps, doubling the current 6.4Gbps of DDR5 RDIMM memory and improving pin speed.

In JEDEC's vision, DDR5 MRDIMM will utilize the same pins, SPD, PMIC, and other designs as existing DDR5 DIMMs, be compatible with the RDIMM platform, and leverage the existing LRDIMM ecosystem for design and testing.

JEDEC stated that these two new technical specifications are expected to bring a new round of technological innovation to the memory market.

- **Micron's MRDIMM DDR5 to Start Mass Shipment in 2H24**

In March 2023, AMD announced at the Memcom 2023 event that it is collaborating with JEDEC to develop a new DDR5 MRDIMM standard memory, targeting a transfer rate of up to 17600 MT/s. According to [a report from Tom's Hardware at that time](#), the first generation of DDR5 MRDIMM aims for a rate of 8800 MT/s, which will gradually increase, with the second generation set to reach 12800 MT/s, and the third generation to 17600 MT/s.

MRDIMM, short for "Multiplexed Rank DIMM," integrates two DDR5 DIMMs into one, thereby providing double the data transfer rate while allowing access to two ranks.

On [July 16, memory giant Micron announced the launch of the new MRDIMM DDR5](#), which is currently sampling and will provide ultra-large capacity, ultra-high bandwidth, and ultra-low latency for AI and HPC applications. Mass shipment is set to begin in the second half of 2024.



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MRDIMM offers the highest bandwidth, largest capacity, lowest latency, and better performance per watt. Micron said that it outperforms current TSV RDIMM in accelerating memory-intensive virtualization multi-tenant, HPC, and AI data center workloads.

Compared to traditional RDIMM DDR5, MRDIMM DDR5 can achieve an effective memory bandwidth increase of up to 39%, a bus efficiency improvement of over 15%, and a latency reduction of up to 40%.

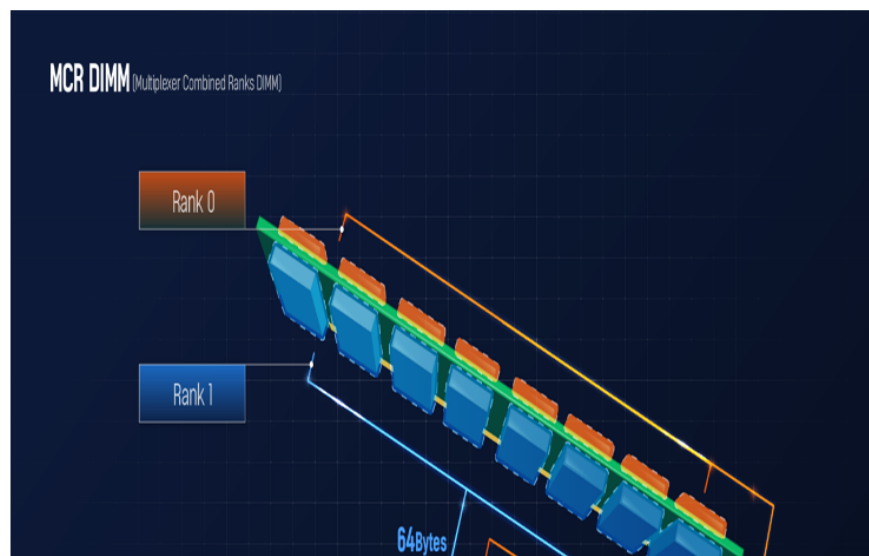
MRDIMM supports capacity options ranging from 32GB to 256GB, covering both standard and high-form-factor (TFF) specifications, suitable for high-performance 1U and 2U servers. The 256GB TFF MRDIMM outruns TSV RDIMM with similar capacity by 35% in performance.

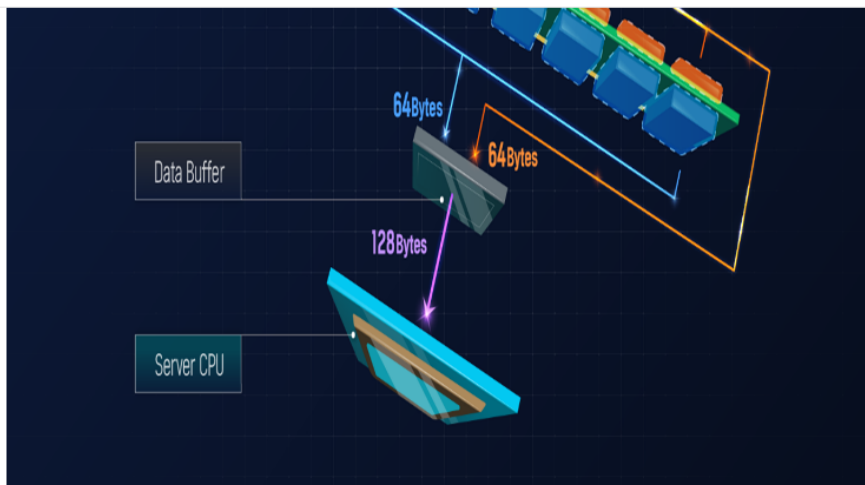
This new memory product is the first generation of Micron's MRDIMM series and will be compatible with Intel Xeon processors. Micron stated that subsequent generations of MRDIMM products will continue to offer 45% higher single-channel memory bandwidth compared to their RDIMM counterparts.

- **SK hynix to Launch MCRDIMM Products in 2H24**

As one of the world's largest memory manufacturers, SK hynix already introduced a product similar to MRDIMM, called MCRDIMM, even before AMD and JEDEC.

MCRDIMM, short for "Multiplexer Combined Ranks² Dual In-line Memory Module," is a module product that combines multiple DRAMs on a substrate, operating the module's two basic information processing units, Rank, simultaneously.





Source: [SK hynix](#)

In late 2022, SK hynix partnered with Intel and Renesas to develop the DDR5 MCR DIMM, which became the fastest server DRAM product in the industry at the time. As per Chinese IC design company Montage Technology's 2023 annual report, MCRDIMM can also be considered the first generation of MRDIMM.

Traditional DRAM modules can only transfer 64 bytes of data to the CPU at a time, while SK hynix's MCRDIMM module can transfer 128 bytes by running two memory ranks simultaneously. This increase in the amount of data transferred to the CPU each time boosts the data transfer speed to over 8Gbps, doubling that of a single DRAM.

At that time, SK hynix anticipated that the market for MCR DIMM would gradually open up, driven by the demand for increased memory bandwidth in HPC. According to SK hynix's FY2024 Q2 financial report, the company will launch 32Gb DDR5 DRAM for servers and MCRDIMM products for HPC in 2H24.

• MRDIMM Boasts a Brilliant Future

MCRDIMM/MRDIMM adopts the DDR5 LRDIMM "1+10" architecture, requiring one MRCD chip and ten MDB chips. Conceptually, MCRDIMM/MRDIMM allows parallel access to two ranks within the same DIMM, increasing the capacity and bandwidth of the DIMM module by a large margin.

Compared to RDIMM, MCRDIMM/MRDIMM can offer higher bandwidth while maintaining good compatibility with the existing mature RDIMM ecosystem. Additionally, MCRDIMM/MRDIMM is expected to enable much higher overall server performance and lower total cost of ownership (TCO) for enterprises.

MRDIMM and MCRDIMM both fall under the category of DRAM memory modules, which have different application scenarios relative to HBM as they have their own independent market space. As an industry-standard packaged memory, HBM can achieve higher bandwidth and energy efficiency in a given capacity with a smaller size. However, due to high cost, small capacity, and lack of scalability, its application is limited to a few fields. Thus, from an industry

have their own independent market space. As an industry-standard packaged memory, HBM can achieve higher bandwidth and energy efficiency in a given capacity with a smaller size. However, due to high cost, small capacity, and lack of scalability, its application is limited to a few fields. Thus, from an industry perspective, memory module is the mainstream solution for large capacity, cost-effectiveness, and scalable memory.

Montage Technology believes that, based on its high bandwidth and large capacity advantages, MRDIMM is likely to become the preferred main memory solution for future AI and HPC. As per JEDEC's plan, the future new high-bandwidth memory modules for servers, MRDIMM, will support even higher memory bandwidth, further matching the bandwidth demands of HPC and AI application scenarios.

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(Photo credit: [SK hynix](#))



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